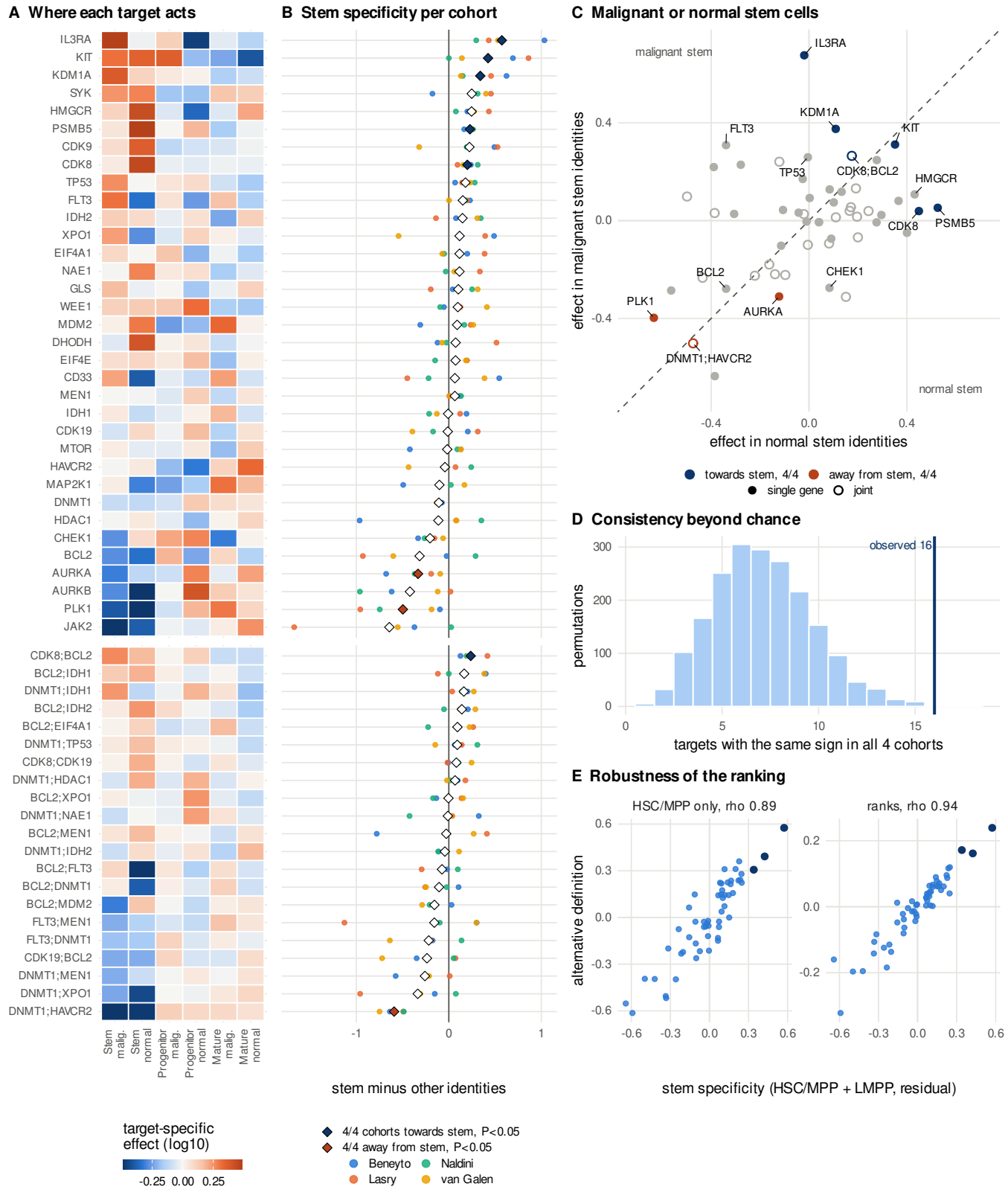

A Where each target acts
IL3RA
KIT
KDM1A
SYK
HMGCR
PSMB5
CDK9
CDK8
TP53
FLT3
IDH2
XPO1
EIF4A1
NAE1
GLS
WEE1
MDM2
DHODH
EIF4E
CD33
MEN1
IDH1
CDK19
MTOR
HAVCR2
MAP2K1
DNMT1
HDAC1
CHEK1
BCL2
AURKA
AURKB
PLK1
JAK2
CDK8;BCL2
BCL2;IDH1
DNMT1;IDH1
BCL2;IDH2
BCL2;EIF4A1
DNMT1;TP53
CDK8;CDK19
DNMT1;HDAC1
BCL2;XPO1
DNMT1;NAE1
BCL2;MEN1
DNMT1;IDH2
BCL2;FLT3
BCL2;DNMT1
BCL2;MDM2
FLT3;MEN1
FLT3;DNMT1
CDK19;BCL2
DNMT1;MEN1
DNMT1;XPO1
DNMT1;HAVCR2
Stem malg.
Stem normal
Progenitor malg.
Progenitor normal
Mature malg.
Mature normal
target-specific effect (log10)
-0.25 0.00 0.25
B Stem specificity per cohort
-1
0
1
stem minus other identities
4/4 cohorts towards stem, P<0.05
4/4 away from stem, P<0.05
Beneyto
Naldini
Lasry
van Galen
C Malignant or normal stem cells
malignant stem
normal stem
IL3RA
KDM1A
FLT3
KIT
TP53
CDK8;BCL2
HMGCR
PSMB5
CDK8
BCL2
CHEK1
PLK1
AURKA
DNMT1;HAVCR2
0.4
0.0
-0.4
effect in malignant stem identities
effect in normal stem identities
towards stem, 4/4
away from stem, 4/4
single gene
joint
D Consistency beyond chance
observed 16
300
200
100
0
permutations
0
5
10
15
targets with the same sign in all 4 cohorts
E Robustness of the ranking
HSC/MPP only, rho 0.89
ranks, rho 0.94
0.6
0.3
0.0
-0.3
-0.6
0.2
0.0
-0.2
alternative definition
stem specificity (HSC/MPP + LMPP, residual)